Microeconomics Analysis: Dynamic Optimization

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Plans

- We start with the basics of **dynamic optimization**
  - discrete time only
  - a bit informal
- Then we are going to apply dynamic optimization techniques to **search problems**
- Finally, we are going look at applications of search theory

References

- Dynamic Optimization
  - Stockey and Lucas, "Recursive Methods in Economic Dynamics", 1989
- Search
  - Lippman and McCall, "The Economics of Job Search: a Survey", *Economic Inquiry*, 1976, pp. 155-189.

Introduction

The main focus is on how our decisions today impact the future outcomes:

- consumption versus investment
  - the more cake you eat today the less is left for tomorrow
  - higher consumption today $\Rightarrow$ lower investment $\Rightarrow$ lower consumption in the future
- inventories
  - having large inventories is costly, but there are economies on delivery and enough product in case of surprisingly high demand
- search
  - reject a job offer today $\Rightarrow$ unemployed tomorrow, but perhaps a better offer will come
  - reject a product $\Rightarrow$ more time spent on search, but perhaps a better product or lower price will be found

Taxonomy

- Deterministic (see chapter 4 of SL)
  - finite horizon
  - infinite horizon
- Stochastic (see chapter 9 of SL)
  - finite horizon
  - infinite horizon

Deterministic Setup

- Time $t = 0, 1, \ldots, T$, where $T \in \mathbb{N} \cup \{\infty\}$
- State $x_t \in X \subseteq \mathbb{R}^n$, where X is a set of possible states
- Actions (control):
$$a_t \in A(x_t)$$
chosen from the set of admissible actions in state x_t
- State dynamics/evolution:
$$x_{t+1} = f(x_t, a_t)$$
- Initial condition: $x_0 \in X$
- Instantaneous/flow payoff: $v(x_t, a_t)$
- Discount factor: δ
- Optimization problem:
$$\sup_{\{a_t\}_{t=0}^{T}} \sum_{t=0}^{T} \delta^t v(x_t, a_t)$$

Finite T: Example

Finite time horizon case can be solved without relying on any theory – just use backward induction.

Problem

You have a cake which can be eaten at most in three days. Your utility from consuming share c_t of the cake on day t is $u = \ln c_t$, and the discount factor equals to δ. What is the optimal strategy of eating the cake?

- Time: $t = 0, 1, 2$
- State: x_t – how much of the cake has left.
- Control: consumption c_t and admissible set $A(x_t) = \{c_t \leq x_t\}$
- State evolution: $x_{t+1} = x_t - c_t$
- Initial condition: $x_0 = 1$
- Problem:

$$\max_{c_0, c_1, c_2} [\ln c_0 + \delta \ln c_1 + \delta^2 \ln c_2]$$

Finite T: Backward Induction I

- Period 2: $c_2 = x_2$, $V_2(x_2) = \ln(x_2)$
- Period 1:
  - now we need to take into account period-2 consumption:
  $$c_2 = x_2 = x_1 - c_1$$
  - utility from period 1 onwards is
  $$V_1 = \ln c_1 + \delta V_2(x_2) = \ln c_1 + \delta \ln x_2 = \ln(x_1 - x_2) + \delta \ln x_2$$
  - Optimizing with respect to state x_2 (or control c_1)
  $$\frac{\partial V_1}{\partial x_2} = -\frac{1}{x_1 - x_2} + \frac{\delta}{x_2} = 0$$
  - Optimal consumption:
  $$c_1 = \frac{x_1}{1+\delta}, \quad c_2 = x_2 = \frac{\delta x_1}{1+\delta}, \quad V_1(x_1) = \ln \frac{x_1}{1+\delta} + \delta \ln \frac{\delta x_1}{1+\delta}$$

Finite T: Backward Induction II

- Period 0:
  - now we need to take into account periods 1 and 2 consumption
  - we know that if we arrive to period-1 with x_1 of our cake, our utility given the optimal behaviour further on is $V_1(x_1)$
  - thus, we have

$$V_0 = \ln c_0 + \delta V_1(x_1) = \ln(x_0 - x_1) + \delta \ln \frac{x_1}{1+\delta} + \delta^2 \ln \frac{\delta x_1}{1+\delta}$$

  - Optimizing with respect to x_1

$$\frac{\partial V_0}{\partial x_1} = \frac{-1}{1 - x_1} + \delta \frac{1}{x_1} + \delta^2 \frac{1}{x_1} = 0$$

  - Optimal solution:

$$x_1 = \frac{\delta(1+\delta)}{1+\delta+\delta^2}, \quad x_2 = \frac{\delta x_1}{1+\delta} = \frac{\delta^2}{1+\delta+\delta^2}$$

$$V_0(1) = \ln \frac{1}{1+\delta+\delta^2} + \delta \ln \frac{\delta}{1+\delta+\delta^2} + \delta^2 \ln \frac{\delta^2}{1+\delta+\delta^2}$$

Finite T: Conclusions

- Solve the problem using backward induction
- On each step we can view the continuation payoff as a function of state variable $V(x_t)$ (time included in x)
- We can reformulate the problem as the choice of optimal sequence of state variables $\{x_t\}_{t=1}^{T}$

General T: Preliminaries

Equipped with the previous knowledge we make some adjustments to the problem:

- we reformulate the choice as the choice over sequences $\{x_t\}_{t=1}^{T}$
- dynamics is restricted to:

$$x_{t+1} \in \Gamma(x_t) = \{f(x_t, a_t) : a_t \in A(x_t)\}$$

- redefine instantaneous pay-off as

$$u(x_t, x_{t+1}) = \max_{a_t \in A(x_t) : f(x_t, a_t) = x_{t+1}} v(x_t, a_t)$$

General T: Sequence Problem

Sequence Problem: Find $V(x_0)$ such that

$$V(x_0) = \sup_{\{x_t\}_{t=1}^{T}} \sum_{t=0}^{T} \delta^t u(x_t, x_{t+1})$$

$$\textbf{s.t. } x_{t+1} \in \Gamma(x_t), \qquad x_0$$

General T: Sequence Problem, Examples

- Infinite cake problem
- Optimal growth problem:
  - Infinite horizon
  - Consumption utility $u = \ln c_t$
  - Production $y_t = k_t^\alpha$
  - Part is consumed, part is reinvested: $c_t = y_t - k_{t+1}$
  - This yields $\Gamma(k_t) = [0, k_t^\alpha]$
  - $$V(k_0) = \sup_{\{k_t\}_{t=1}^{\infty}} \sum_{t=0}^{\infty} \delta^t \ln(k_t^\alpha - k_{t+1})$$

    s.t. $k_{t+1} \in \Gamma(x_t)$

Bellman Equation

Bellman equation expresses the value function as a combination of a **flow payoff** and a discounted **continuation payoff**:

$$V(x_t) = \sup_{x_{t+1} \in \Gamma(x_t)} \left[u(x_t, x_{t+1}) + \delta V(x_{t+1}) \right]$$

- flow payoff $u(x_t, x_{t+1})$
- continuation payoff $V(x_{t+1})$ – future value function

Claim

$V(x_t)$ is a solution to Sequence Problem if and only if it is a solution to Bellman Equation

Some regularity conditions are required, see Assumptions 4.1 and 4.2 and Theorems 4.2 and 4.3 in SL book for formal argument

Sketch of the Proof I

SP⇒BE

$$
\begin{aligned}
V(x_0) &= \sup_{x_{t+1}\in\Gamma(x_t)} \sum_{t=0}^{T} \delta^t u(x_t, x_{t+1}) \\
&= \sup_{x_{t+1}\in\Gamma(x_t)} \left[u(x_0, x_1) + \sum_{t=1}^{T} \delta^t u(x_t, x_{t+1}) \right] \\
&= \sup_{x_{t+1}\in\Gamma(x_t)} \left[u(x_0, x_1) + \delta \sum_{t=1}^{T} \delta^{t-1} u(x_t, x_{t+1}) \right] \\
&= \sup_{x_1\in\Gamma(x_0)} \left[u(x_0, x_1) + \delta \sup_{x_{t+1}\in\Gamma(x_t)} \sum_{t=1}^{T} \delta^{t-1} u(x_t, x_{t+1}) \right] \\
&= \sup_{x_1\in\Gamma(x_0)} [u(x_0, x_1) + \delta V(x_1)]
\end{aligned}
$$

Sketch of the Proof II

BE$\Rightarrow$SP

$$
\begin{aligned}
V(x_0) &= \sup_{x_1 \in \Gamma(x_0)} [u(x_0, x_1) + \delta V(x_1)] \\
&= \sup_{x_{t+1} \in \Gamma(x_t),\ t=1,2} \left[u(x_0, x_1) + \delta u(x_1, x_2) + \delta^2 V(x_2)\right] \\
&\vdots \\
&= \sup_{x_{t+1} \in \Gamma(x_t)} \left[u(x_0, x_1) + \ldots + \delta^n u(x_n, x_{n+1}) + \delta^{n+1} V(x_{n+1})\right] \\
&= \sup_{x_{t+1} \in \Gamma(x_t)} \sum_{t=0}^{T} \delta^t u(x_t, x_{t+1})
\end{aligned}
$$

Sufficient condition for existence is $\lim_{t \to \infty} \delta^t V(x_t) = 0$ for all feasible sequences.

Solving Problems

Now we have the tool for solving the general type of problem – Bellman Equation

How to apply it?

In general, you need to solve a difference (differential if in continuous time) equation

Equivalent: iterate functional operator

Shortcut: guess a solution

- typically guess the functional form
- then solve for coefficients

Sometimes finite problem give useful insights

Cake Problem: Setup

Problem

You have a cake which can be eaten in your lifetime. Your utility from consuming share c_t of the cake on day t is $u = \ln c_t$, and the discount factor equals to δ. What is the optimal strategy of eating the cake?

So, we are solving

$$\max_{c_t} \sum_{t=0}^{\infty} \delta^t \ln c_t \qquad \text{s.t. } x_t - c_t = x_{t+1}$$

Rewriting it as a Sequence Problem:

$$V(x_0) = \max_{x_t} \sum_{t=0}^{\infty} \delta^t \ln(x_t - x_{t+1}) \qquad \text{s.t. } x_{t+1} \leq x_t$$

Corresponding Bellman Equation:

$$V(x_t) = \max_{x_{t+1} \leq x_t} [\ln(x_t - x_{t+1}) + \delta V(x_{t+1})]$$

Cake Problem: Guess

$$V(x_t) = \max_{x_{t+1} \le x_t} [\ln(x_t - x_{t+1}) + \delta V(x_{t+1})]$$

How does V look like?

Recall our finite cases:

$$V_2(x_2) = \ln x_2$$

$$V_1(x_1) = \ln \frac{x_1}{1+\delta} + \delta \ln \frac{\delta x_1}{1+\delta} = (1+\delta) \ln x_1 + [\delta \ln \delta - (1+\delta) \ln(1+\delta)]$$

$$V_0(x_0) = \ln \frac{x_0}{1+\delta} + \delta \ln \frac{\delta x_0}{1+\delta+\delta^2} + \delta^2 \ln \frac{\delta^2 x_0}{1+\delta+\delta^2}$$

Conjecture:

$$V(x) = C_0 + C_1 \ln(x)$$

Cake Problem: Solution I

Bellman Equation

$$C_0 + C_1 \ln(x_t) = \max_{x_{t+1}} [\ln(x_t - x_{t+1}) + \delta C_0 + \delta C_1 \ln x_{t+1}]$$

F.O.C.:

$$\frac{-1}{x_t - x_{t+1}} + \frac{\delta C_1}{x_{t+1}} = 0 \;\Rightarrow\; x_{t+1} = \frac{\delta C_1}{1 + \delta C_1} x_t$$

Plug back into Bellman Equation:

$$C_0 + C_1 \ln(x_t) = \ln\left(x_t - \frac{\delta C_1}{1 + \delta C_1} x_t\right) + \delta\left[C_0 + C_1 \ln\left(\frac{\delta C_1}{1 + \delta C_1} x_t\right)\right]$$

Now it is simple but tedious: compute free coefficient and coefficient in front of $\ln$, get two equations with two unknowns

Cake Problem: Solution II

$$
\begin{aligned}
C_0 + C_1 \ln x_t = {} & (1 + \delta C_1) \ln x_t \\
& + \delta C_0 - (1 + C_1) \ln(1 + \delta C_1) + C_1 \ln(\delta C_1)
\end{aligned}
$$

Thus,

$$
C_1 = 1 + \delta C_1 \;\Rightarrow\; C_1 = \frac{1}{1-\delta}
$$

$$
C_0 = \frac{1}{1-\delta} \left(\frac{\delta}{1-\delta} \ln \delta - \ln(1-\delta) \right)
$$

Check the algebra!!

We are more interested in the optimal path.

$$
x_{t+1} = \frac{\delta C_1}{1 + \delta C_1} x_t = \delta x_t
$$

Optimal Growth Problem: Setup

- Infinite horizon
- Consumption utility $u = \ln c_t$
- Production $y_t = k_t^\alpha$, $0 < \alpha < 1$
- Part is consumed, part is reinvested: $c_t = y_t - k_{t+1}$
- Initial condition: k_0

Corresponding Bellman Equation:

$$V(k_t) = \max_{k_{t+1}} [\ln[k_t^\alpha - k_{t+1}] + \delta V(k_{t+1})]$$

Optimal Growth Problem: Guess

Try two-period problem:

$$V_1(k_1) = \ln k_1^\alpha$$

$$V_0 = \ln[k_0^\alpha - k_1] + \delta \ln k_1^\alpha$$

F.O.C.:

$$\frac{-1}{k_0^\alpha - k_1} + \frac{\alpha\delta}{k_1} = 0 \Rightarrow k_1 = \frac{\alpha\delta}{1+\alpha\delta} k_0^\alpha$$

Plug it back into value function:

$$V_0(k_0) = \ln\left(\frac{1}{1+\alpha\delta} k_0^\alpha\right) + \delta \ln\left(\frac{\alpha\delta}{1+\alpha\delta} k_0^\alpha\right)$$

So, again conjecture $V(k) = C_0 + C_1 \ln k$

Optimal Growth Problem: Solution I

Bellman Equation

$$C_0 + C_1 \ln(k_t) = \max_{k_{t+1}} \left[\ln(k_t^\alpha - k_{t+1}) + \delta C_0 + \delta C_1 \ln k_{t+1}\right]$$

F.O.C.:

$$\frac{-1}{k_t^\alpha - k_{t+1}} + \frac{\delta C_1}{k_{t+1}} = 0 \;\Rightarrow\; k_{t+1} = \frac{\delta C_1}{1 + \delta C_1} k_t^\alpha$$

Plug back into Bellman Equation:

$$C_0 + C_1 \ln(k_t) = \ln\left(k_t^\alpha - \frac{\delta C_1}{1 + \delta C_1} k_t^\alpha\right) + \delta\left[C_0 + C_1 \ln\left(\frac{\delta C_1}{1 + \delta C_1} k_t^\alpha\right)\right]$$

Now follow the same steps as before...

Optimal Growth Problem: Solution II

$$
\begin{aligned}
C_0 + C_1 \ln k_t = {} & \alpha(1+\delta C_1) \ln k_t \\
& + \delta C_0 - (1+C_1)\ln(1+\delta C_1) + C_1 \ln(\delta C_1)
\end{aligned}
$$

Thus,

$$
C_1 = \alpha(1+\delta C_1) \;\Rightarrow\; C_1 = \frac{\alpha}{1-\alpha\delta}
$$

Obtain the optimal path:

$$
k_{t+1} = \frac{\delta C_1}{1+\delta C_1} k_t^{\alpha} = \alpha\delta k_t^{\alpha}
$$

Envelope Theorem

Theorem

$$\frac{\partial V(x_t)}{\partial x_t} = \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t}$$

Proof

$$\begin{aligned}
\frac{\partial V(x_t)}{\partial x_t} &= \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t} + \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_{t+1}^*}\frac{dx_{t+1}^*}{dx_t} + \delta\frac{\partial V(x_{t+1}^*)}{\partial x_{t+1}^*}\frac{dx_{t+1}^*}{dx_t} \\
&= \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t} + \left[\frac{\partial u(x_t, x_{t+1}^*)}{\partial x_{t+1}^*} + \delta\frac{\partial V(x_{t+1}^*)}{\partial x_{t+1}^*}\right]\frac{dx_{t+1}^*}{dx_t} \\
&= \frac{\partial u(x_t, x_{t+1}^*)}{\partial x_t}
\end{aligned}$$

Discrete Problems: Towers of Hanoi

Bellman Equation can be applied to discrete problems as well.

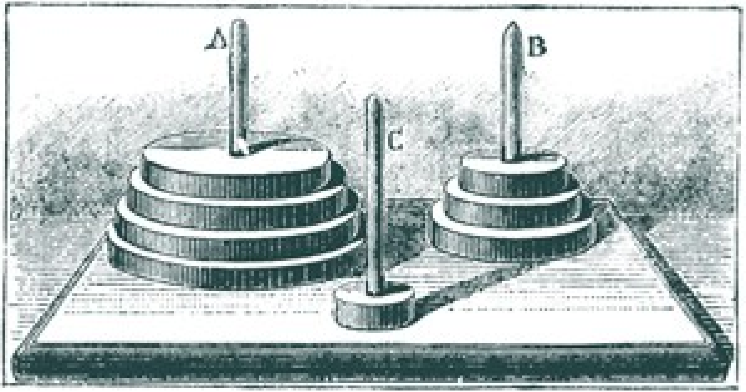
Problem

In a temple in Hanoi monks are displacing 64 golden disks with a whole in the middle, all of different sizes, on three ivory sticks. They observe the following rules:

1. *is is only allowed to move one disk at a time from one stick to another one*
2. *it is forbidden to put a disk on top of a smaller disk*

If monks move one disk per second and follow the optimal strategy, how long will it take to move all disks from one stick to another?

Towers of Hanoi



Towers of Hanoi: Solution

We have minimization problem rather than maximization, but otherwise it is very similar.

Try to come up with **Bellman Equation**

Let $V(n)$ be the minimum number of moves required to move n disks from one stick to another

Then, in order to move $n+1$ disks we need:

- move top n disks to another stick – $V(n)$ moves
- move the largest disk – one move
- put n disks back on top of the large one – $V(n)$ moves

Thus, our Bellman Equation is

$$V(n+1) = 2V(n) + 1$$

So, $V(n) = 2^n - 1$ and it will take $2^{64} - 1$ seconds, roughly 585 billion years...

Stochastic Optimization

This can be very complicated...

The whole probability distribution over sequences may depend on current action in a rather complicated way

Usually consider Markov processes, i.e. when the probability distribution over future state variable depends on current state variable and control only

We are going to restrict our attention to search problems